

Junseo **Lee**

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Education

B.S. in Naval Architecture and Ocean Engineering Mar. 2022 – Feb. 2028 (Expected)
Seoul, Korea

Seoul National University (SNU)

• GPA: 3.35/4.30 (Major : 3.44/4.30)
• Relevant coursework: Structural Mechanics, Structural Vibrations, Finite Element Theory, and Ship Design

Research Experiments

Undergraduate Intern Aug. 2026 – Present

OST Lab. / Seoul National University / Advisor : Prof. Dokyun Kim

- AI-based double hull tanker midship optimisation tool development
- Multi-Fidelity Fluid-Structure interaction analysis

Undergraduate Visiting Researcher Aug. 2025 – Feb. 2026

MORIN Lab. / KAIST / Advisor : Prof. Jinwhan Kim

- Organized existing maritime sensor datasets for model development and evaluation, and participated in field experiments at Jeongok Port.
- LiDAR-based maritime vessel Re-Identification (ReID). : Developed CIEF-Net, a neural network that integrates local curvature features with spatial coordinates through early fusion for vessel re-identification.
- Cross-View Representation Learning under Partial Observations. : Developed a cross-view masked autoencoder to learn vessel representations by reconstructing a target view from partial observations of another viewpoint.

Publications & Presentations

Poster Presentation | KROS2026 (PyeongChang) Feb. 2026

"Curvature-Integrated Early Fusion Network for LiDAR-Based Marine Vessel Re-Identification."Junseo Lee, Dongwook Lee, and Jinwhan Kim.

Poster Presentation | UR 2026 (Osaka) Jul. 2026

"Maritime Vessel Re-Identification under Partial Observations Using Cross-View Masked Autoencoder." Junseo Lee, Dongwook Lee, Sol Han, and Jinwhan Kim.

Awards & Honors

DSME Prize, Autonmous Ship Contest for Freshman(SNU) Jun. 2022

Developing Colision Detection Algorithm

2nd Prize, Ocean Engineering CAE contest(KSOE) Sep. 2026

Team Leader, LLM-based midship optimisation software development

Skills

Programming	Python, MATLAB, C/C++
Machine Learning	PyTorch, Open3D
Language	Korean(Native), English(Fluent)

Military Service

DMZ East corridor patrol Feb. 2024 – Aug. 2025

Sergeant discharged

- Commendation for Successful Defection Inducement Operation (CJCS, Aug. 2024).
- Certificate of Achievement (UNCMAC-S, Aug. 2025).

Research Interests

My research interests lie at the intersection of machine learning and engineering, with a focus on integrating physical principles and domain knowledge to support reliable analysis, interpretable predictions, and informed design decisions. I am particularly interested in self-supervised learning of 3D geometric representations for maritime perception that is robust to partial observations, viewpoint changes, and limited real-world data. I also seek to explore simulation-informed learning and multi-fidelity modeling for structural analysis and optimization, balancing computational efficiency with physical understanding and engineering constraints.